G L Bajaj Group of Institutions, Mathura

Worksheet Problems on Data Structures and Algorithms

Worksheet 5

Q1:Count the triplets

Givenanarrayarr.Thetaskistocountallthetripletssuchthatthesumof two elements equals the third element.

Examples:

Input:arr[]=[1,5,3,2] Output:2

Explanation: There are 2 triplets: 1 + 2 = 3 and 3 +2 = 5

Input: arr[] = [2,3,4] Output: 0

Explanation: No such triplet exits

Q2: Missing in Array

Youaregivenanarrayarrofsizen-1thatcontainsdistinctintegers intherangefrom1ton(inclusive).This arrayrepresentsapermutation oftheintegersfrom1tonwithoneelementmissing.Yourtaskisto identify and return the missing element.

Examples:

Input:arr[]=[1,2,3,5] Output:4

Explanation: All the numbers from 1 to 5 are present except 4.

Input: arr[] = [8,2,4,5,3,7,1] Output: 6

Explanation: All the numbers from 1 to 8 are present except 6.

Input: arr[] = [1] Output:
2

Explanation: Only 1 is present so the missing element is 2.

Q3: Merge Without Extra Space

Given two sorted arrays a[] and b[] in non-decreasing order. Merge them in sorted order without using any extra space. Modify a so that it contains the first n elements and modify b so that it contains the last m elements.

Examples:

Input: a[] = [1, 3, 5, 7], b[] = [0, 2, 6, 8, 9]

Output: a[] = [0, 1, 2, 3], b[] = [5, 6, 7, 8, 9]

Explanation: After merging the two non-decreasing arrays, we get, 0 1 2 3 5 6 7 8 9

Input: a[] = [10, 12], b[] = [5, 18, 20]

Output: a[] = [5, 10], b[] = [12, 18, 20]

Explanation: After merging two sorted arrays we get 5 10 12 18 20.

Input: a[] = [0, 1], b[] = [2, 3]

Output: a[] = [0, 1], b[] = [2, 3]

Explanation: After merging two sorted arrays we get 0 1 2 3.

Q4: Rearrange Array Alternately

Given a sorted array of positive integers. Your task is to rearrange the array elements alternatively i.e first element should be max value, second should be min value, third should be second max, fourth should be second min and so on.
Note: Modify the original array itself. Do it without using any extra space. You do not have to return anything.

Example 1:

Input:
n = 6

arr[] = {1,2,3,4,5,6}

Output: 6 1 5 2 4 3

Explanation: Max element = 6, min = 1, second max = 5, second min = 2, and so on... Modified array is : 6 1 5 2 4 3.

Example 2:

Input:

n = 11

arr[]={10,20,30,40,50,60,70,80,90,100,110}

Output:110 10 100 20 90 30 80 40 70 50 60

Explanation: Max element = 110, min = 10, second max = 100, second min = 20, and so on... Modified array is :

110 10 100 20 90 30 80 40 70 50 60.

Q5:Number of pairs

Giventwopositiveintegerarraysarrandbrr,findthenumberofpairssuch that xy > yx (raised to power of) wherexisanelementfromarrandyisan elementfrombrr.

Examples :

Input: arr[] = [2, 1, 6], brr[] = [1, 5] Output: 3

Explanation: The pairs which follow xy > yx are: 21 > 12, 25 > 52 and 61 > 16 .

Input: arr[] = [2 3 4 5], brr[] = [1 2 3] Output: 5

Explanation: The pairs which follow xy > yx are: 21 > 12 , 31 > 13 , 32 > 23 , 41 > 14 , 51 > 15 .